

Adaptive Online Non-stochastic Control

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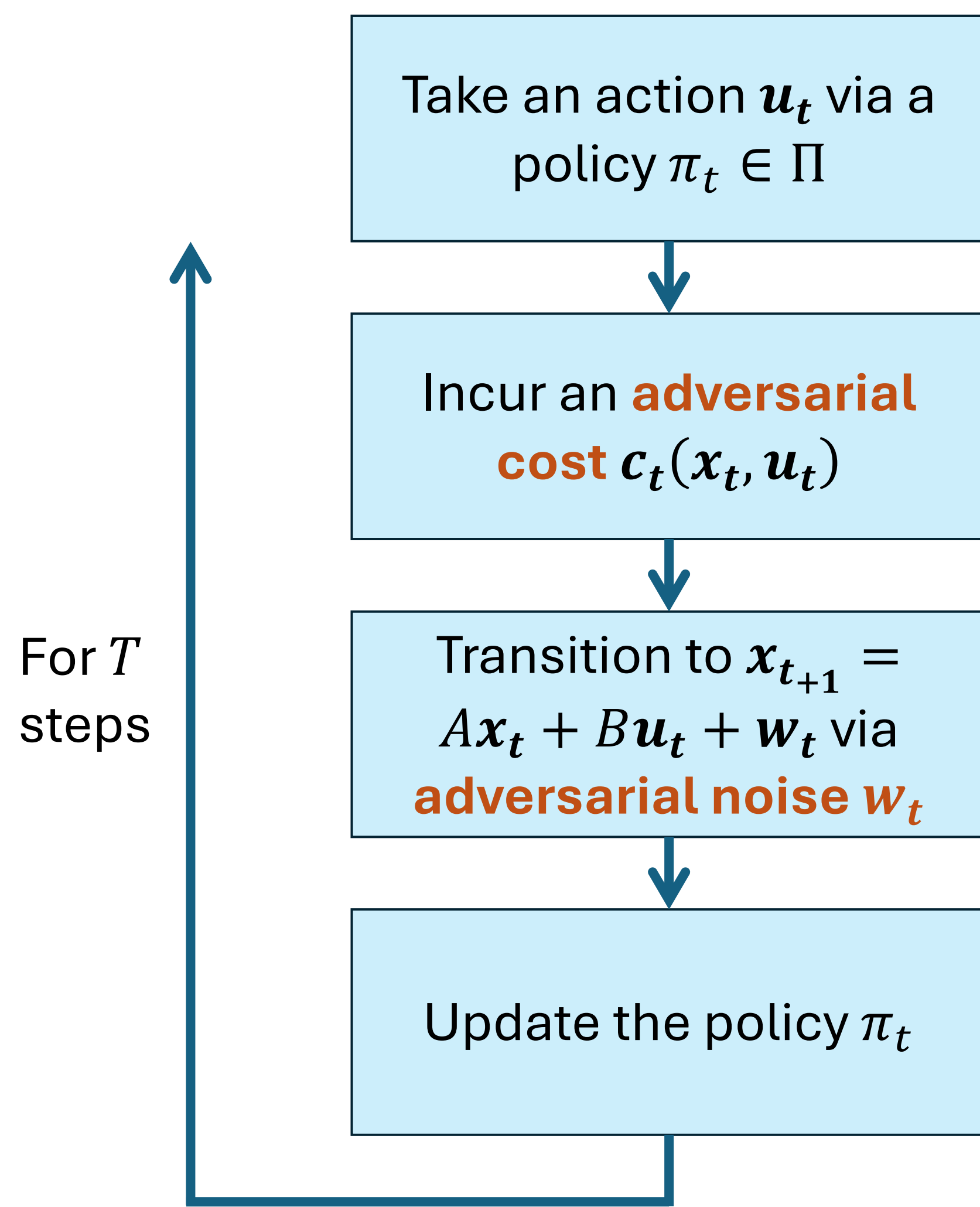


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1

Problem Setup

- Control a **Non-stochastic** time-slotted system:



- Target: **best policy in Π**

$$\pi^* = \operatorname{argmin}_{\pi \in \Pi} \sum_{t=1}^T c_t(x_t(\pi), u_t(\pi))$$

$$\text{s.t., } x_{t+1} = Ax_t + Bu_t + w_t, \quad \forall t.$$

- Metric: **policy regret**

$$\mathcal{R}_T = \sum_{t=1}^T c_t(x_t, u_t) - c_t(x_t(\pi^*), u_t(\pi^*))$$

Where:

- x_t, u_t : The *actual* trajectory due to $\{\pi_t\}_T$.
- $x_t(\pi^*), u_t(\pi^*)$ the *counterfactual* due to π^* .

- Features:

- π^* is based on **witnessed** costs and noise.
- $\mathcal{R}_T = o(T) \Rightarrow \pi_t$ **achieves** π^* 's performance.

- The Standard Controller: **Gradient P.C.** :

- Execute $u_t = \sum_{j=1}^p M_t^{[j]} w_{t-j}$ (DAC)
- Update Policy Param: $M_{t+1} = M_t + \eta g_t$,
 $g_t = \nabla_M c_t(x_t(\pi), u_t(\pi))$

- GPC 's Policy Regret:

$$\mathcal{R}_T = \mathcal{O}(g\sqrt{T})$$

Polynomials in
matrix norms

**Universal Upper
Bound** on $\|g_t\|$

- What if $\|g_t\|$ is small (easy environments)?

RQ:

Can the bound on $\mathcal{R}_T$ be modulated by $\|g_t\|$ instead of g ?

2

AdaFTRL-C Algorithm & Guarantee

- At each slot t :

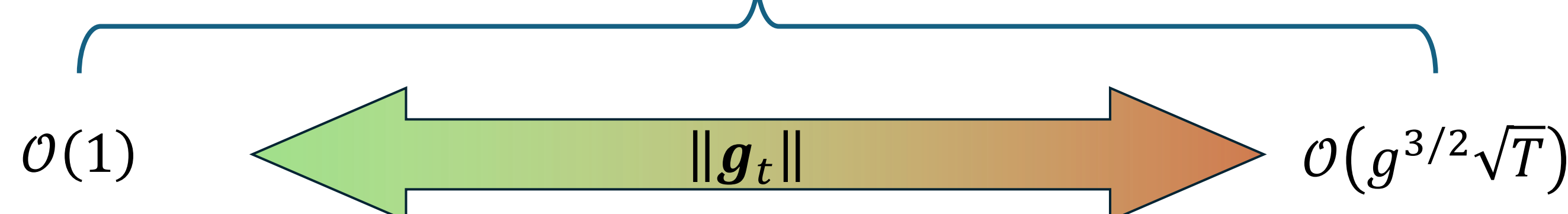
- Execute $u_t = \sum_{j=1}^p M_t^{[j]} w_{t-j}$ (DAC class)
- Set regularization parameter $\lambda_t \propto \sqrt{\sum_{s=1}^{t-1} \|g_s\|^2}$
- Update policy params:

$$M_{t+1} = \operatorname{argmin}_{\pi \in \Pi} \{c_t(x_t(\pi), u_t(\pi)) + \lambda_t \|M\|^2\}$$

Adaptive regularization

- Theorem:

$$\mathcal{R}_T = \mathcal{O}\left(\sqrt{g} \sqrt{\sum_{t=1}^T \|g_t\|^2}\right)$$



3

AdaFTRL-C Proof Concepts

Step 1: $c_t(x_t, u_t) = c_t(x_t(\pi_t), u_t(\pi_t)) + \text{residual}$ (New NSC+FTRL result)

- Lemma:

$$\nu_t \doteq \left| c_t(x_t, u_t) - c_t(x_t(\pi_t), u_t(\pi_t)) \right| \leq l z \sum_{i=0}^{t-1} (1-\delta)^i \sum_{s=t-i-1}^{t-1} \frac{g_s}{\sigma \sqrt{g_{1:s}}} \doteq \hat{\nu}_t.$$

Step 2: $c_t(x_t(\pi_t), u_t(\pi_t))$ has a sublinear regret (Standard FTRL result)

- Lemma:

$$\sum_{t=1}^T c_t(x_t(\pi_t), u_t(\pi_t)) - c_t(x_t(\pi^*), u_t(\pi^*)) \approx \mathcal{O}(\sqrt{\sum_t \|g_t\|^2})$$

Step 3: The residual is *also* sublinear (New result)

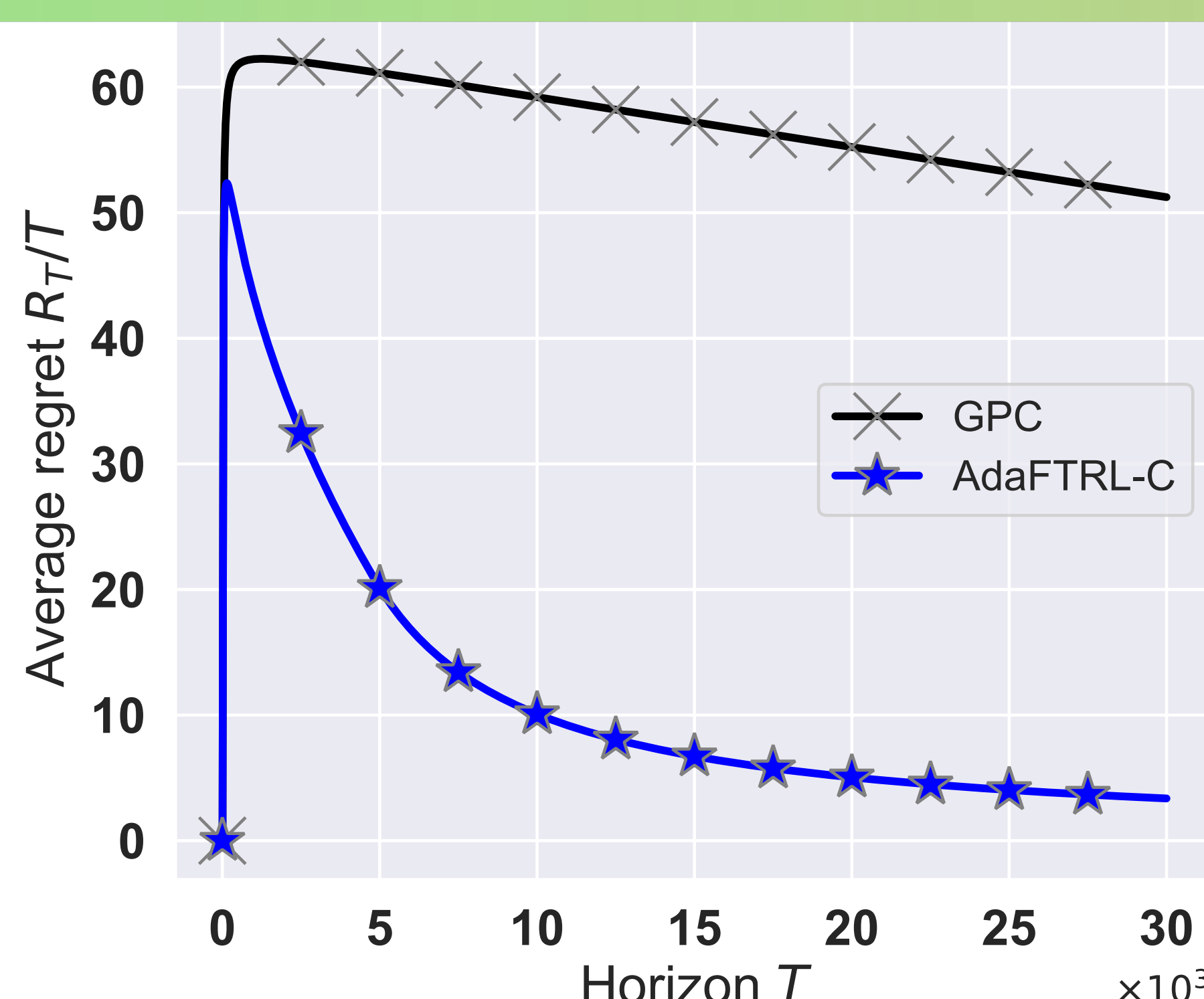
- Lemma:

$$\sum_{t=1}^T \hat{\nu}_t \approx \mathcal{O}(\sqrt{\sum_t \|g_t\|^2})$$

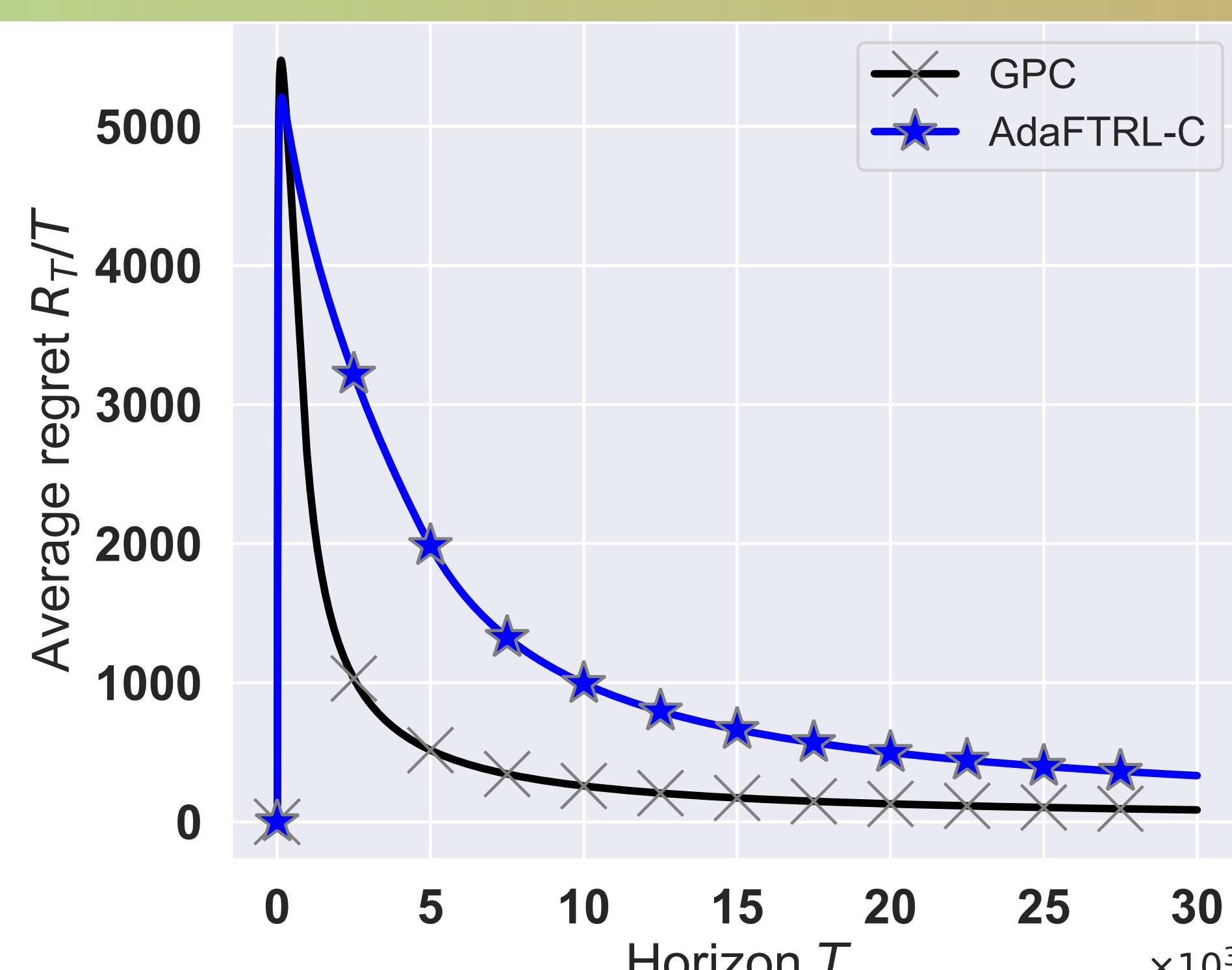
4

Numerical Examples

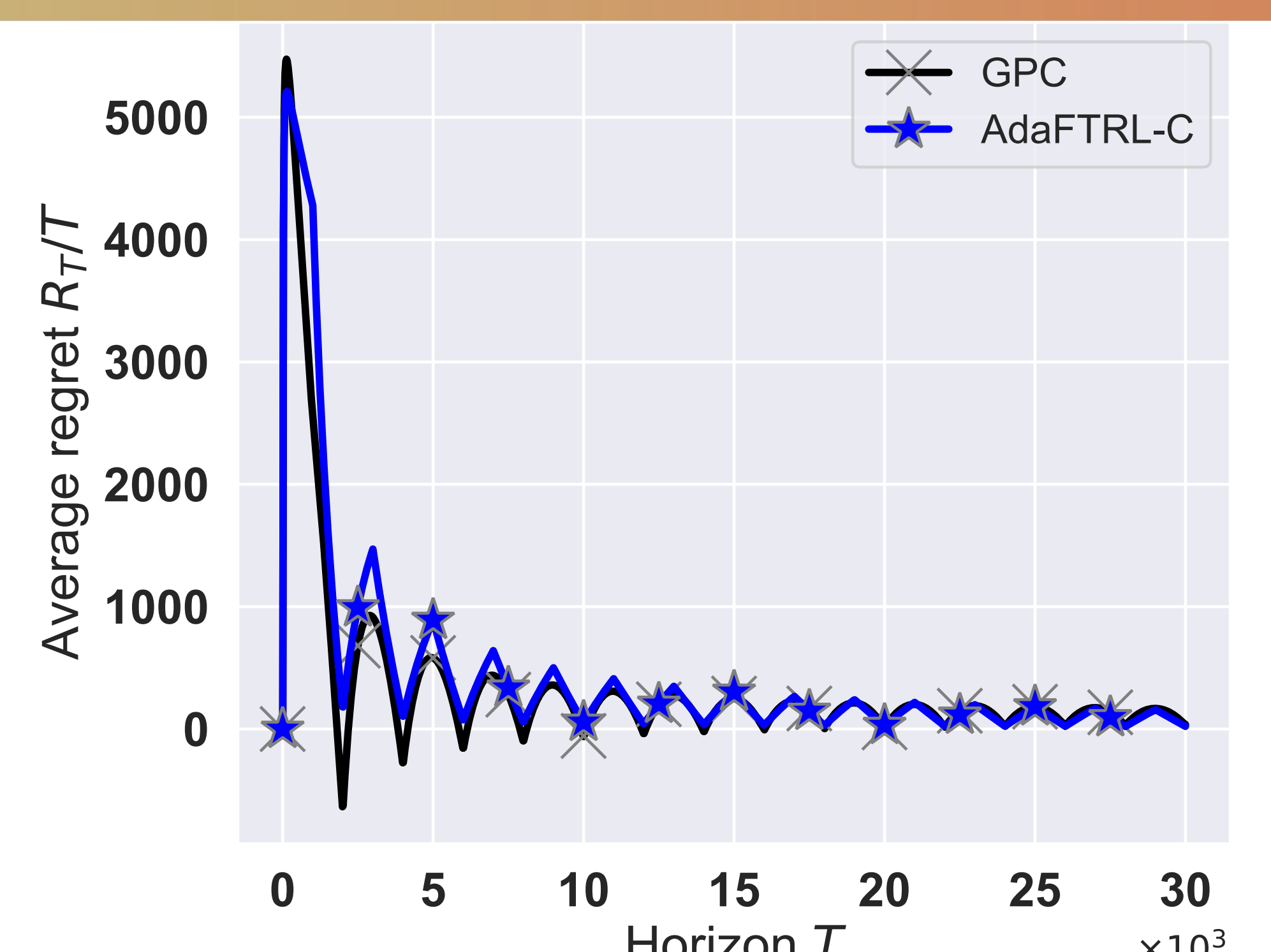
Disturbances & costs are **10% their max**:



Disturbances & costs are **max**:



Disturbances & costs are **max & alternating**:



References:

- NSC: Hazan et. al. "The non-stochastic control problem". In Proc. of ALT, 2020.
- FTRL & Adaptive OCO: McMahan. "A survey of algorithms and analysis for adaptive online learning". J. Mach. Learn. Res., 2017.
- More details in the paper: Mhaisen et. al. "Adaptive online non-stochastic control" In Proc. of L4DC, 2024.